ANALYSTLAB AFRICA

Machine Learning Internship Programme · Week 2

Data Preprocessing Report

ShopSmart Retail Group — Preparing the Data for Machine Learning

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1. Business objective

Two supervised-learning problems are planned: predicting the value of an order (regression, target Order_Value) and predicting whether an order is returned (classification, target Is_Returned). This report documents every step that turns nine raw ShopSmart tables into one machine-learning-ready dataset, and why each decision was taken. It accompanies the Jupyter notebook ShopSmart_Week2_Preprocessing.ipynb.

2. Data inspection

All nine tables were loaded and inspected for shape, data types, missing values, duplicates and descriptive statistics. The modelling table has 50,000 orders (one row per order) and an overall return rate of 4.88%.

Check	Result
Missing cells (all 9 tables)	0
Exact duplicate rows (all 9 tables)	0
Foreign keys that fail to resolve	0 (customers, products, stores, returns all match)
Order period	1 Jan 2025 – 31 Dec 2025
Date columns stored as text	Order, registration, store opening, return and restock dates

3. Data-quality issues

Because the tables are complete, the audit tested business logic rather than only counting blanks. Thirteen issues were found:

Issue	Table	Affected	Treatment
Line_Total subtracts Discount as a flat number (not as a %)	order_items	50,000 (100%)	Target is a deterministic formula of its inputs (leakage); discount effect is negligible
Order Unit_Price differs from catalogue Selling_Price (>1 unit)	orders vs products	50,000 (100%)	Two conflicting price sources; order price used
Returns recorded on Cancelled/Pending orders	returns	1,685 (67.4%)	Impossible business flow; Order_Status kept but treated with caution
Return_Date earlier than Order_Date	returns	1,252 (50.1%)	Return timing is unreliable; not used as a feature

Issue	Table	Affected	Treatment
Orders with more than one return record	returns	58 (2.3%)	Collapsed to one order-level flag
Refund_Amount larger than the order's Line_Total	returns	270 (10.8%)	Refund amount not used (post-event, inconsistent)
Order placed before the customer's registration date	orders vs customers	5,159 (10.3%)	Tenure would be negative; clipped to 0 and flagged
Duplicate customer e-mail addresses	customers	49 (1%)	E-mail is personal data; dropped
Duplicate customer phone numbers	customers	4 (0.1%)	Phone is personal data; dropped (formats are also inconsistent)
States that appear under more than one Region	stores	6 (60%)	Store_Region is unreliable; kept as a weak feature
Store names that are duplicated	stores	15 (60%)	Store_ID used as the key, name dropped
Sub_Category holds a single constant value ('General')	products	500 (100%)	Zero information; dropped
Suppliers with malformed phone numbers (<7 digits)	suppliers	80 (16.4%)	Table not used in the model

Figure 1 — Data-quality issues found by the logic audit

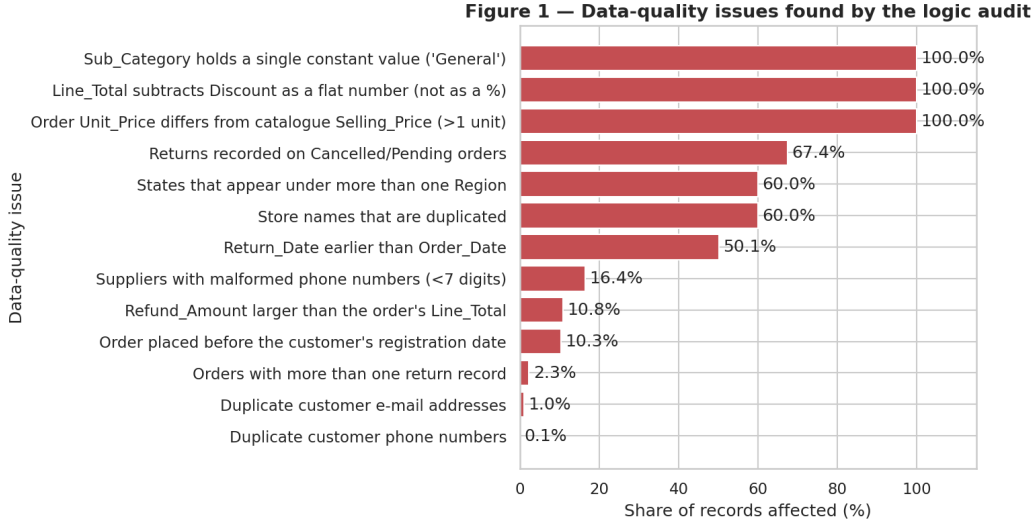


Figure 1 — Data-quality issues found by the logic audit

Most important finding: Line_Total equals Quantity × Unit_Price − Discount, where Discount is a 0–20 whole number. The target is therefore a near-exact function of its inputs, which explains the R^2 of 1.0 in Week 1. Treating Discount as a percentage would lower the average order value by about 15,000.

4. Data cleaning decisions

Step	Decision	Reason
Missing values	No imputation needed	None exist; plan (median / mode / flag) documented for future refreshes
Duplicates	None removed; 58 orders with two return records collapsed to one flag	No exact duplicates; the order is the unit of prediction

Step	Decision	Reason
Text values	Stripped and standardised case	Levels were already consistent; protects future loads
Data types	Dates parsed with explicit month/day/year format; categories typed	Prevents silent day/month swaps; enables time features
Validation	Assertions on quantity 1–5, discount 0–20, prices > 0, age 18–100	Fails loudly if bad values appear
Joins	Orders joined to items, customers, products, stores, returns with 1:1 or many-to-1 validation	A bad key raises an error instead of duplicating rows
Target	Is_Returned = order has at least one return record	Matches Week 1 (4.88%)
Leakage	Return count, return date and refund amount not carried forward	Only known after the outcome

5. Feature engineering

Fourteen features were created, three columns were renamed for clarity, and columns without modelling value were removed.

Feature	Built from	Why it was created
Order_Month, Order_Quarter, Order_DayOfWeek, Is_Weekend	Order date	Expose seasonality and weekday behaviour; models cannot use raw dates
Gross_Value	Quantity × Unit_Price	Basket size before discount (also the source of the leakage)
Shipping_Share	Shipping ÷ Gross_Value	Shipping burden relative to basket size
Unit_Margin_Pct	Catalogue selling and cost price	Product profitability
Price_vs_Catalog_Gap	Order price − catalogue price	Captures the conflicting price sources
Customer_Tenure_Days, Ordered_Before_Registration	Order and registration dates	Customer age on the platform; flag keeps impossible rows visible (negative tenure set to 0)
Age_Band	Customer age	Ordered age groups: 18-29, 30-44, 45-59, 60+
Prior_Orders	Running count per customer	Purchase history using only earlier orders (no future leakage)
Store_Age_Years, Is_Local_Store	Store opening date; customer vs store state	Store maturity; whether customer bought in home state

Renamed: Line_Total → Order_Value, Discount → Discount_Pct, Age → Customer_Age. **Removed:** first name, last name, e-mail, phone (personal data); Customer_City (duplicates state); Sub_Category (constant); Brand and Product_Name (about 500 unique values, effectively IDs); all ID columns except the row key Order_ID.

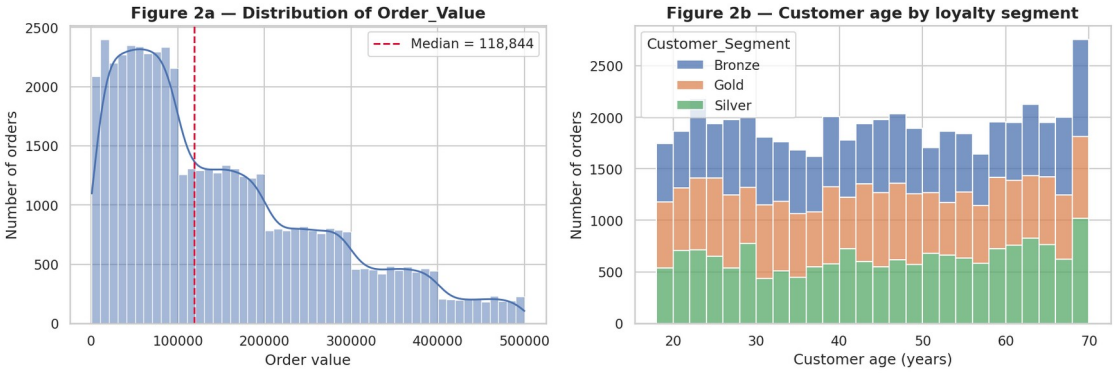


Figure 2 — Order_Value distribution and customer age by segment

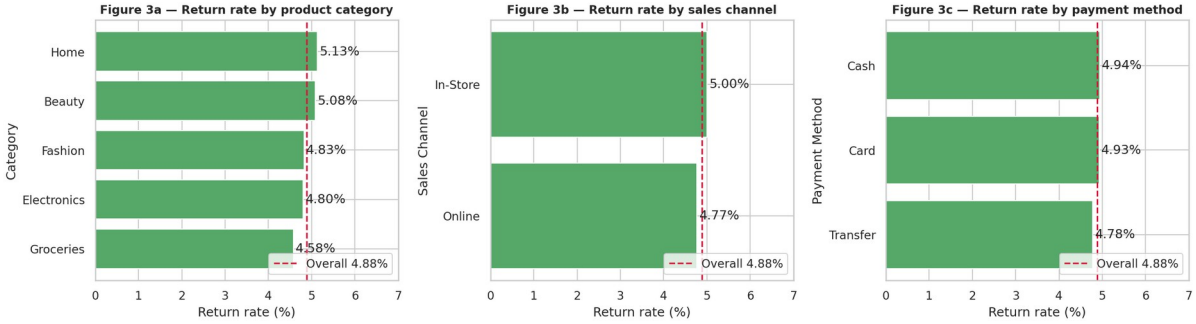


Figure 3 — Return rate by category, channel and payment method

6. Encoding and scaling

6.1 Train/test split

The data was split before any statistics were learned: 40,000 training rows and 10,000 test rows (80/20, stratified on Is_Returned, random_state 42). The return rate is 4.88% in both parts. All scalers were fitted on the training rows only, so the test set stays unseen.

6.2 Encoding

Technique	Applied to	Reason
Ordinal	Customer_Segment (Bronze < Silver < Gold); Age_Band	Categories have a natural order that the numbers preserve
Label (0/1)	Sales_Channel, Gender	Two levels only; one column suffices, no false ordering
One-hot (first level dropped)	Payment_Method, Order_Status, Category, Customer_State, Store_Region	Nominal categories; avoids implying order; drop-first avoids collinearity

6.3 Scaling

Scaler choice follows a rule applied to each column in the training data, rather than one scaler for everything:

Scaler	Rule	Columns
RobustScaler	|skew| > 0.5	Gross_Value, Prior_Orders, Shipping_Share
MinMaxScaler	≤ 12 distinct values (bounded)	Age_Band_Ord, Customer_Segment_Ord, Discount_Pct, Order_DayOfWeek, Order_Month, Order_Quarter, Quantity
StandardScaler	Symmetric continuous	Cost_Price, Customer_Age, Customer_Tenure_Days,

Scaler	Rule	Columns
		Price_vs_Catalog_Gap, Selling_Price, Shipping_Cost, Store_Age_Years, Unit_Margin_Pct, Unit_Price

Binary and one-hot columns stay as 0/1. Targets are never scaled. Scaling changes units, not shape (Figure 4), so large-number columns such as Gross_Value (up to about 500,000) no longer dominate small ones such as Quantity (1–5).

Figure 5 — Effect of the three scalers (training rows)

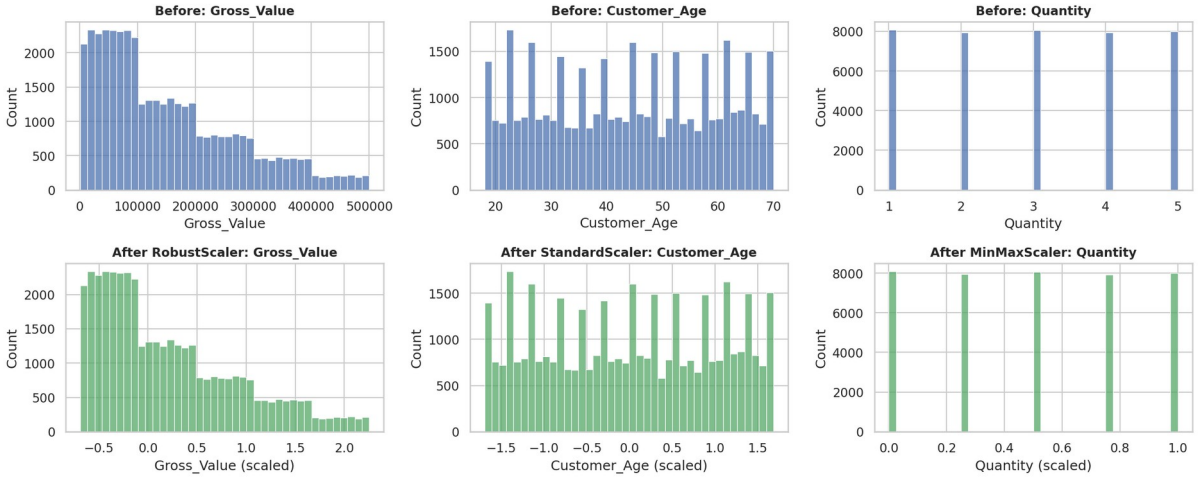


Figure 4 — Effect of the three scalers (training rows)

7. Outlier detection

Outliers were measured on original values using the IQR rule (1.5 × IQR fences), Z-scores (|z| > 3) and box plots.

Feature	Max	Skew	IQR outliers	Z > 3
Gross_Value	499975	0.89	410 (0.82%)	0
Order_Value	499960	0.89	410 (0.82%)	0
Shipping_Cost	5000	0	0 (0%)	0
Unit_Price	99998	-0.01	0 (0%)	0
Customer_Age	70	-0.01	0 (0%)	0
Customer_Tenure_Days	1089	0.19	0 (0%)	0
Prior_Orders	23	0.64	557 (1.11%)	170
Store_Age_Years	5.92	0	0 (0%)	0
Shipping_Share	4.54	9.79	5428 (10.86%)	800

- **Gross_Value / Order_Value:** 410 baskets (0.82%) flagged by IQR but none beyond 3 standard deviations; they are the top of a legitimate right-skewed range (maximum about 500,000 = price 100,000 × quantity 5).
- **Prior_Orders:** 557 values (1.1%) are loyal repeat customers (up to 23 earlier orders).
- **Shipping_Share:** extreme ratios arise from very small baskets; removed later by the variance threshold.

Decision: all rows were kept. Deleting or capping high baskets would remove the most valuable customers and bias value predictions downward. Robust scaling limits their influence instead.

Figure 6 — Box plots by return outcome (points beyond the whiskers are IQR outliers)

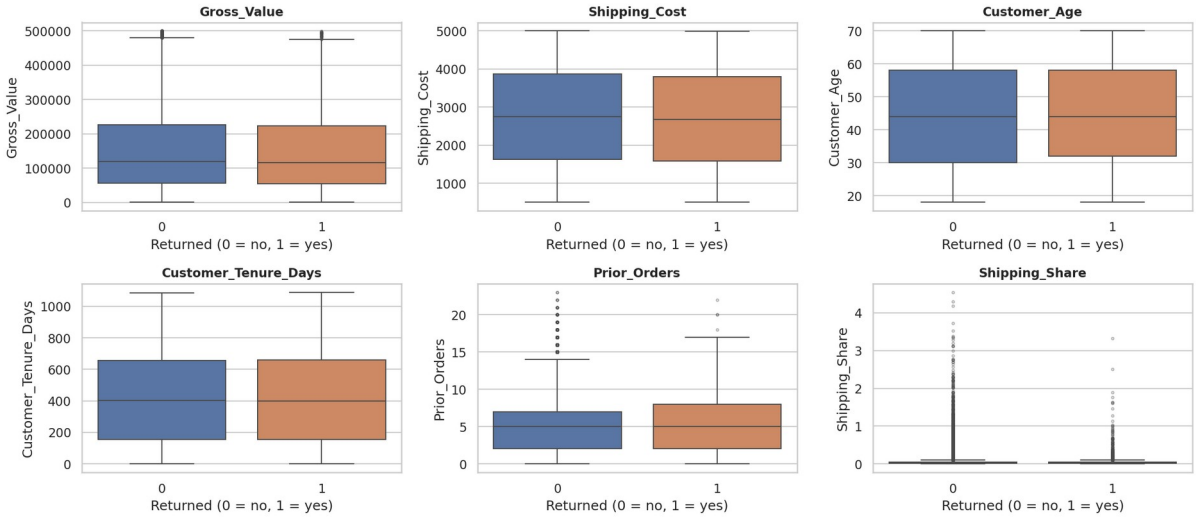


Figure 5 — Box plots by return outcome

8. Feature selection

Starting from 44 encoded and scaled candidates, four methods were applied to the training rows only:

- **Variance threshold (0.01):** removed Shipping_Share (almost constant after rescaling because of its extreme tail).
- **Correlation analysis (|r| > 0.85):** removed Order_Month, Customer_Age, Selling_Price, each a near-duplicate of a retained feature (e.g. Order_Month vs Order_Quarter, Customer_Age vs Age_Band).
- **Leakage check:** Gross_Value correlates with Order_Value at 1.0000, so it is excluded from the regression feature set.
- **Importance versus a random-noise probe:** only 5 of 39 features exceeded a column of pure noise, and only slightly; cross-validated ROC-AUC is 0.4955 ± 0.0116.

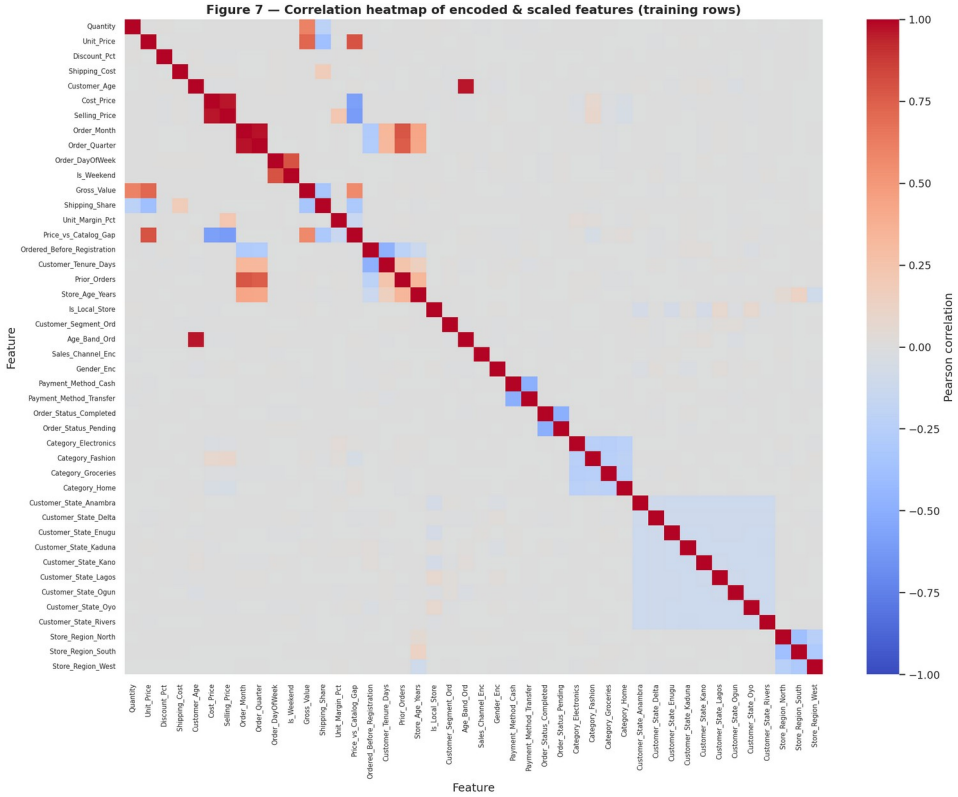


Figure 6 — Correlation heatmap of encoded and scaled features

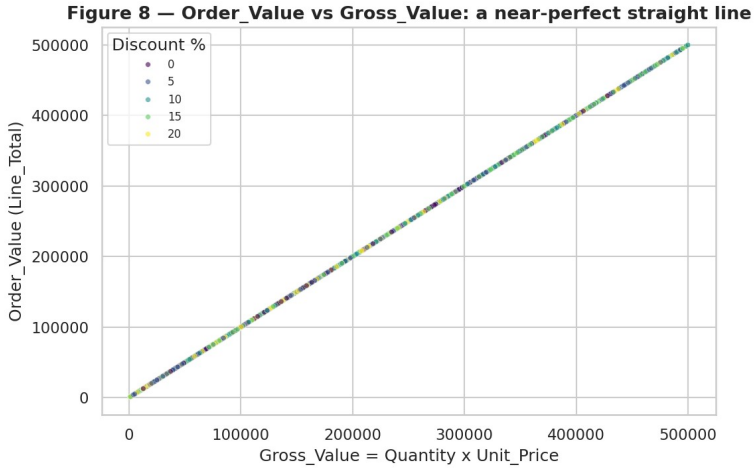


Figure 7 — Order_Value against Gross_Value: a near-perfect straight line

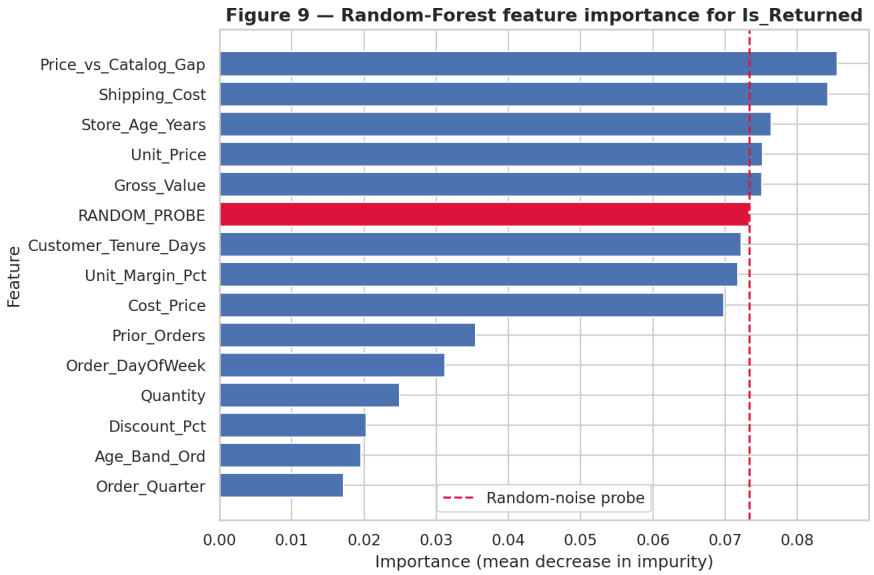


Figure 8 — Random-Forest importance for Is_Returned, with random-noise probe

Result: 40 features were retained. Three feature sets are saved in feature_sets.json: classification (all 40), regression without Gross_Value, and regression without any arithmetic inputs.

Regression feature set	Test R^2
With Gross_Value (leaky)	1.000
Without Gross_Value (quantity and price kept)	1.000
Without arithmetic inputs (quantity, price, discount, price gap, shipping share)	-0.002

9. Final dataset

File	Shape	Contents
ShopSmart_ML_Ready.csv	50,000 × 44	Order_ID, Split, two targets, 40 encoded and scaled features
ShopSmart_Cleaned.csv	50,000 × 33	Readable cleaned table (no personal data), before encoding and scaling
feature_sets.json, preprocessing_artifacts.joblib	—	Feature lists, fitted encoders and scalers for reuse on new data

Readiness checks passed: no missing values, no infinite values, all feature columns numeric, unique order keys, both train and test partitions present.

10. Key observations

- The raw data is complete but logically flawed; counting blanks alone would have missed every important problem.
- Order_Value is determined by quantity and price, so high regression accuracy reflects arithmetic. This is the cause of the Week 1 result of $R^2 = 1.0$.
- Returns appear unrelated to the recorded product, customer, store, price and timing fields (ROC-AUC 0.4955).

- Only 4.88% of orders are returned, so accuracy is a misleading metric; recall, precision and ROC-AUC should be used.
- Only 33% of orders are Completed; revenue analysis must filter by order status.

11. Limitations and next steps

- Confirm with the data owner whether Discount is a percentage; if so Order_Value should be recomputed.
- Returns information (timing, status, refund) is inconsistent and was used only as a yes/no flag.
- Week 3: train baseline models on the stored split, use class weights or resampling for returns, compare against a majority-class baseline, and report the regression with and without arithmetic inputs.